封凡

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研究方向：面向复杂生物系统的多模态 AI + 结构化知识平台构建

- 生物医疗数据的多模态人工智能
- 糖尿病和胰岛生物学领域的数据挖掘
- 构建集中式知识库与人工智能平台，赋能科学发现
- 针对大规模四维基因组、空间转录组及多组学数据进行数据挖掘

教育经历和科研背景

范德堡大学医学中心 | **博士后研究员 / 数据科学家** | **2024.2 - 至今**
药学院，内分泌与糖尿病系 | **美国，纳什维尔**

导师: Dr. Marcela Brissova

- 主导开发 PanKbase——人类胰腺集中式知识库；并在知识库中应用**知识图谱**
- 开发用于分析人类胰腺**空间蛋白质组/多组学数据**的计算工具
- **多模态人工智能**，包括将大语言模型与智能体人工智能应用于糖尿病研究

密歇根大学 | **生物信息学 博士** | **2018.8 - 2023.12**
计算药物学与生物信息学系 | **美国，安娜堡**

导师: Dr. Jie Liu

- 运用**机器学习模型**分析**染色质三维结构**及表观基因组/基因组特征
- 开发 Genomic Knowledgebase，一个基于**知识图谱**的开源科学平台，用于联合分析人类基因组/表观基因组/转录组/四维基因组数据
- **生物信息学基础模型 (foundation models)**

北京大学 | **化学生物学 学士** | **2014.9 – 2018.7**
化学与分子工程学院 | **中国，北京**

导师: 来鲁华教授

北京大学 | **经济学 双学位学士** | **2015.9 - 2018.7**
国家发展研究院 | **中国，北京**

发表文章 (* 表示共同第一作者)

人工智能+人类遗传学 (主要为三维基因组数据)

- *(Preprint)* Hong, C. K.*, **Feng, F.***, Ramanathan, V., Liu, J., & Hansen, A. S. (2025). Genome structure mapping with high-resolution 3D genomics and deep learning. ***bioRxiv***, 2025-05. (***Nature Genetics*** in revision)
- Impact of Genomic Variation on Function (IGVF) consortium. (2024). Deciphering the impact of genomic variation on function. ***Nature***, 633(8028), 47-57.
- Zhang, Z., **Feng, F.**, Qiu, Y., & Liu., J. (2023). A generalizable framework to comprehensively predict epigenome, chromatin organization, and transcriptome. ***Nucleic Acids Research***, gkad436.
- **Feng, F.**, Yao, Y., Wang, X. Q. D., Zhang, X., & Liu, J. (2022). Connecting high-resolution 3D chromatin

organization with epigenomics. ***Nature Communications***, 13(1), 1-10.

- Zhang, Z., **Feng, F.**, & Liu, J. (2022). Characterizing collaborative transcription regulation with a graph-based deep learning approach. ***PLoS Computational Biology***, 18(6), e1010162.
- **Feng, F.**, Lai, L., & Pei, J. (2018). Computational chemical synthesis analysis and pathway design. ***Frontiers in Chemistry***, 6, 199.

糖尿病领域的多模态数据整合

- Evans-Molina, C.*, Pettway, Y. D.*, Saunders, D. C.*, ..., **Feng, F.**, ... & Brissova, M. (2025). Heterogeneous endocrine cell composition defines human islet functional phenotypes. ***bioRxiv***. (Accepted in Principle by ***Nature Communications***)
- Saunders, D. C.*, **Feng, F.***, Hopkirk, A. E.*, Aamodt, K. I.*, Hart, N.*, Pan, F. C.*, ..., Wright, C. V. E., Powers, A. C. & Brissova, M. (2024). Mapping histologic and functional maturation of human endocrine pancreas across early postnatal periods. ***bioRxiv***, 2024-12. (***Nature Communications*** in revision)

知识库，知识图谱，检索增强生成 **(RAG)**

- **Feng, F.**, Tang, F., Gao, Y., Zhu, D., ... & Liu, J. (2023). GenomicKB: a knowledge graph for the human genome. ***Nucleic Acids Research***, 51.D1, D950-D956.
- *(Preprint)* Huang, Y., Han, Z., ..., **Feng, F.**, ... & Liu, J. (2024). Building a literature knowledge base towards transparent biomedical AI. ***bioRxiv***, 2024-09. (Submitted to ***Nature Communications***)

针对基因组数据（重点关注三维基因组）的严格计算流程与分析

- **Feng, F.***, Moran, S. P.*, & Liu, J. (2025) Statistically rigorous and computationally efficient chromatin stripe detection with Quagga. ***Genome Biology***, 36.1: 159-168
- *(Preprint)* Wang, X. Q. D.*, Yang, W.*, **Feng, F.***, Shum, M., Bell, O., Liu, J., Zhang, X., & Dou, Y. (2025). HOXA9 Couples Transcriptional Control with Higher-Order Chromatin Architecture in Acute Myeloid Leukemia, (Submitted to ***Cancer Discovery***)
- *(Preprint)* Moran, S., Fan, Z., Zanovello, M., **Feng, F.**, Wang, D. X. Q., Lu, C., Liu, J., Zhang, X. (2025). Pan-cancer 3D genomic analysis revealed extremely long Polycomb loops as the biomarker for sensitivity to Polycomb inhibition, ***bioRxiv***, 2025-12 (Submitted to ***Cancer Discovery***)
- Luo, X., Huang, Y., Zeng, H., Tao, Y., Bao, X., **Feng, F.**, Hopkirk, A., ... & Liu, J. (2025). CelLink: integrating single-cell multi-omics data with weak feature linkage and imbalanced cell populations. ***Nucleic Acids Research***, 53.22: gkaf1270.
- Tao, Y., **Feng, F.**, Luo, X., Reihsmann, C. V., Hopkirk, A. L., Cartailler, J. P., ... & Liu, J. (2024). CNTools: A computational toolbox for cellular neighborhood analysis from multiplexed images. ***PLOS Computational Biology***, 20(8), e1012344.
- Varshney, A., Manickam, N., Orchard, P., Tovar, A., Ventresca, C., Zhang, Z., **Feng, F.**, ... & Parker, S. C. (2023). Population-scale skeletal muscle single-nucleus multi-omic profiling reveals extensive context specific genetic regulation. ***bioRxiv***, 2023-12. (Accepted in Principle by ***Nature Genetics***)
- Himadewi, P.*, Wang, X. Q. D.*, **Feng, F.***, Gore, H., Liu, Y., Yu, L., ... & Zhang, X. (2021). 3'HS1 CTCF binding site in human β-globin locus regulates fetal hemoglobin expression. ***Elife***, 10, e70557.
- Li, X., **Feng, F.**, Pu, H., Leung, W. Y., & Liu, J. (2021). scHiCTools: A computational toolbox for analyzing single-cell Hi-C data. ***PLoS Computational Biology***, 17(5), e1008978.
- Zhang, X., Wang, X. Q. D., Gore, H., Himadewi, P., **Feng, F.**, & Liu, J. (2020). 3D Genomics of Acute Meyloid Leukemia Reveals the Imbalance between DNA Methylation Canyon Interactions and Leukemic Specific Enhancer Network Interactions. ***Blood***, 136, 45.

产品与专利

- **GenomicKB –人类基因组知识库 (研究数据库)**
  领导该研究数据库的开发工作。该数据库被列为密歇根大学创新合作页面上的产品，并被**诺和诺德**公司采购用于药物研发。
- **PanKbase –用于糖尿病研究的人类胰腺集中式知识库** (pankbase.org)
  该项目受美国国立卫生研究院下属的国家糖尿病、消化和肾脏疾病研究所资助。作为知识库团队的核心成员参与工作。

学术报告

- "PanKbase – a knowledge base for the pancreas & diabetes community", Sugar Science D-Challenge Seminar, Virtual, Oct. 2025
- "Defining the composition and spatial organization of human pancreatic cells with computational approaches", Beta Cell Interest Group (BIG) Seminar, Nashville, USA, Feb. 2025
- "Defining human pancreatic cell composition and spatial distribution with machine learning methods", AI in Precision Medicine of Diabetes and Other Chronic Diseases Workshop, Rockville, USA, Oct. 2024
- "A generalizable framework to comprehensively predict epigenome, chromatin organization, and transcriptome", Great Lakes Bioinformatics (GLBIO) conference, Montreal, Canada, May 2023
- "An overview of Hi-C downstream analysis", Chromatin 3D Folding Workshop, UT Health, Houston, USA, Mar. 2023
- "GenomicKB: a knowledge graph for human genome", Intellectual Systems of Molecular Biology (ISMB), Lyon, France, Jul. 2023
- "GenomicKB: a knowledge graph for human genome", Tools and Technology Seminar, University of Michigan, Ann Arbor, USA, Jan. 2023
- "Connecting high-resolution 3D chromatin organization with epigenomics", Intellectual Systems of Molecular Biology (ISMB), Madison, USA, Jul. 2022
- "An accurate and interpretable model for predicting high-resolution 3D chromatin organization", Tools and Technology Seminar, University of Michigan, Ann Arbor, USA, Apr. 2020
- "An accurate and interpretable model for predicting high-resolution 3D chromatin organization", 4DN Consortium Joint Analysis Working Group, Virtual, Sep. 2019

教学经历

- 研究生助教/讲师, BIOINF 593/EECS 598: Machine Learning in Computational Biology (计算生物学中的人工智能), University of Michigan, 2022 Fall

指导学生

- Yujuan Fu, University of Michigan, Undergraduate Research Assistant, Apr. 2019 – Apr. 2020
- Shuyuan Yang, University of Michigan, Undergraduate Research Assistant, May 2022 – Apr. 2024
- Yijia Gao, University of Michigan, Undergraduate Research Assistant, May 2022 – Apr. 2024
- Keval Patel, Vanderbilt University, Undergraduate Research Intern, Sep. 2024 – Jun. 2025
- Orianna Bernal-Assis, Vanderbilt University, Summer Research Intern, Jun. 2025 – Aug. 2025
- Kai Lee, Vanderbilt University, Undergraduate Research Intern, Sep. 2025 – Dec. 2025

审稿经历

- Reviewer, Bioinformatics, since 2025
- Co-reviewer, Nature Metabolism, since 2024
- Co-reviewer, Nature Communications, since 2023
- Reviewer, PLoS Computational Biology, since 2022
- Reviewer, Conference on Neural Information Processing Systems (NeurIPS) AI4Science Workshop, since 2021
- Reviewer, Conference on Neural Information Processing Systems (NeurIPS) Generative AI & Biology Workshop, since 2023
- Reviewer, International Conference on Machine Learning (ICML) AI4Science Workshop, since 2022

获得奖项

- Postdoctoral Excellence in Innovation Award, Vanderbilt 2026
- IIDP Data Resource - Trainee Scholar Award (DRTSA) 2025
- AI Workshop Young Scholar Travel Award, National Institute of Diabetes and Digestive and Kidney Disease 2024
- Rackham Conference Travel Award, University of Michigan 2022, 2023
- Program in Biomedical Sciences Fellowship, University of Michigan 2019
- 北京大学“化学之星”毕业生特别奖 2018
- 北京大学明德奖学金 2015
- 四川省理科高考第一名 2014
- 全国高中生化学奥林匹克竞赛（决赛）金牌 2013